

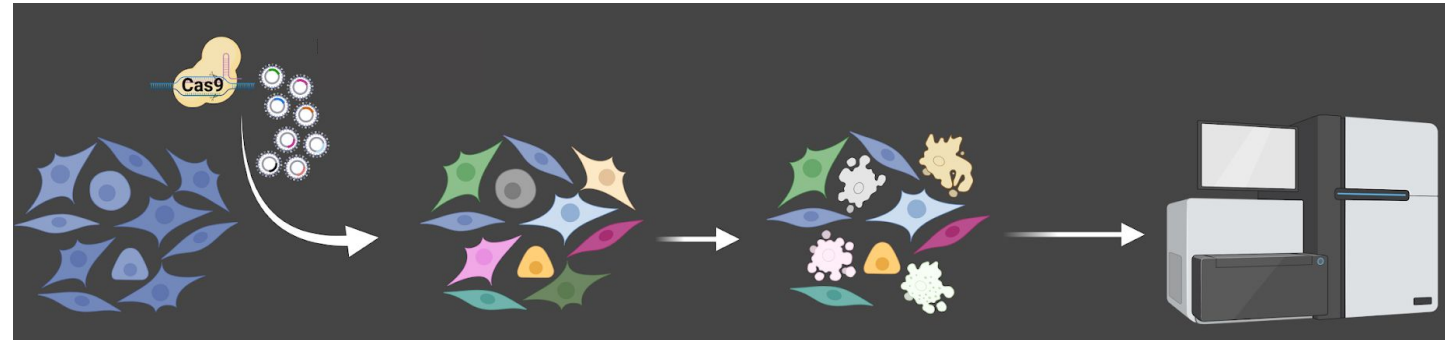


CRISPR screening informatics

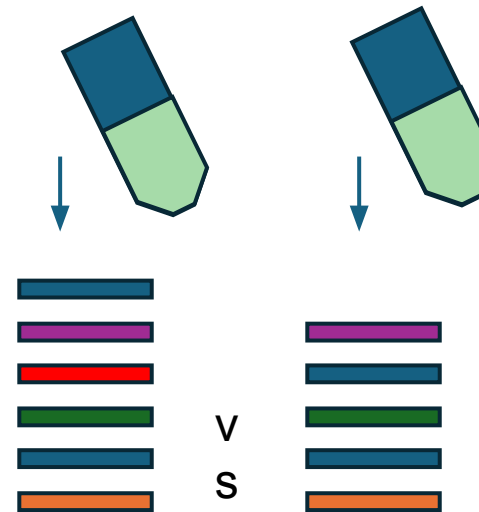
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Principal Bioinformatician
Adams Faculty
30/07/25

CRISPR screening

- Deliver guides into a population of cells expressing Cas9 or a dCas9-fusion.
- Conduct some sort of phenotypic screen.
- Sample from the population at different timepoints.
- Isolate guide RNAs and compare their abundance at different timepoints using NGS.

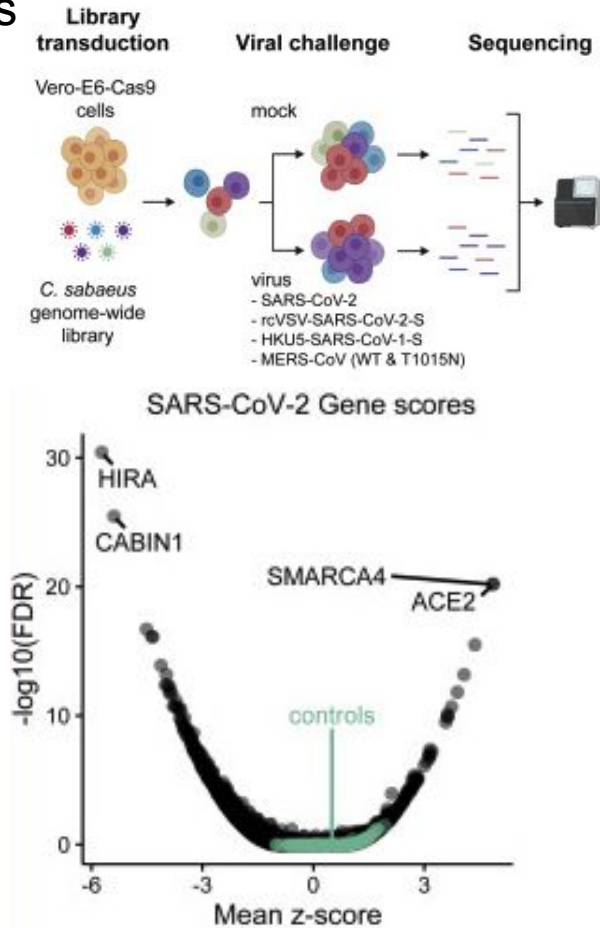


Guide
RNAs



Application of CRISPR screens

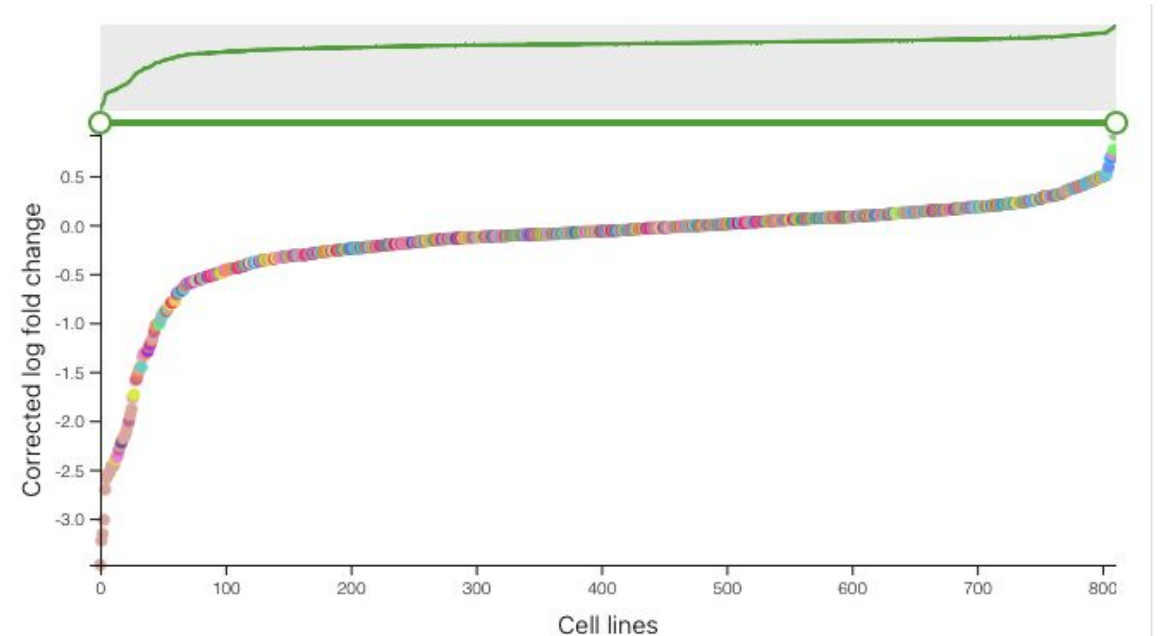
Hypothesis generation -> identify candidate genes



Adapted from Wei, Jin et al. "Genome-wide CRISPR Screens Reveal Host Factors Critical for SARS-CoV-2 Infection." *Cell* vol. 184,1 (2021): 76-91.e13.

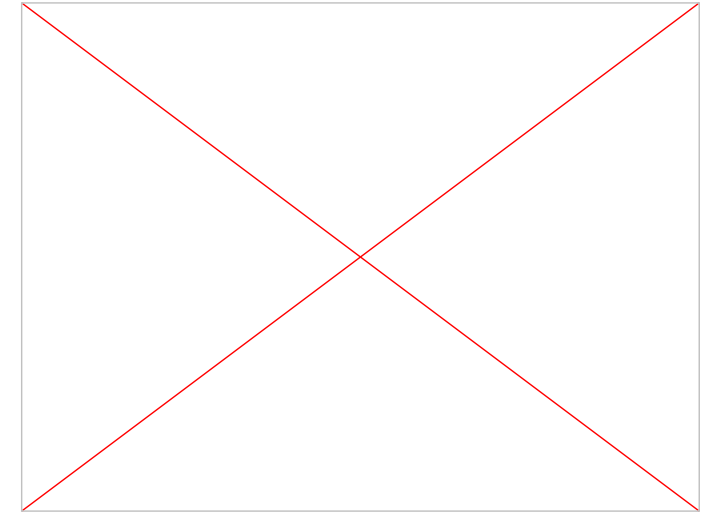
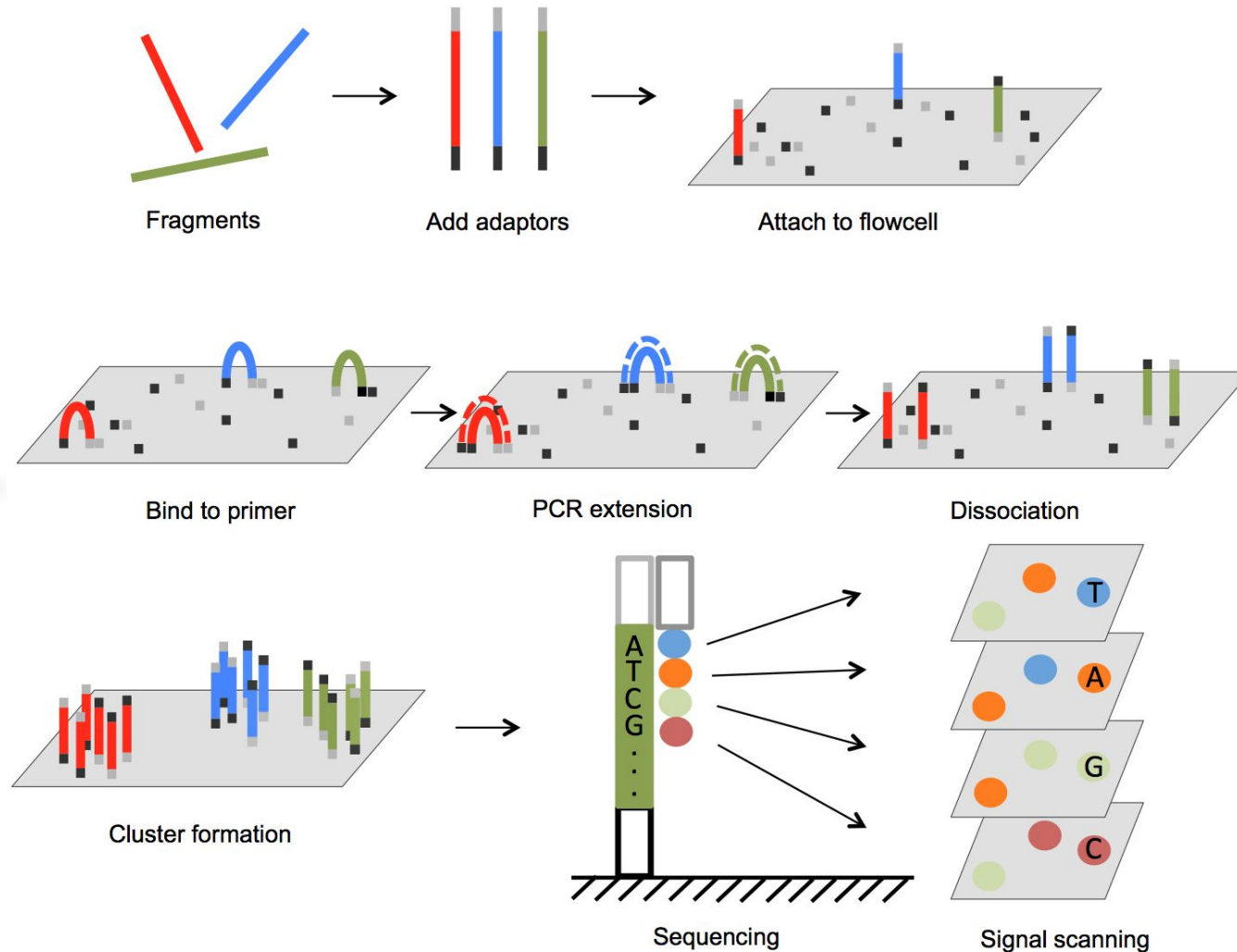
Drug development -> cancer tractability

BRAF Log Fold changes in > 800 cancer cell lines - reflecting sensitivity to knockout



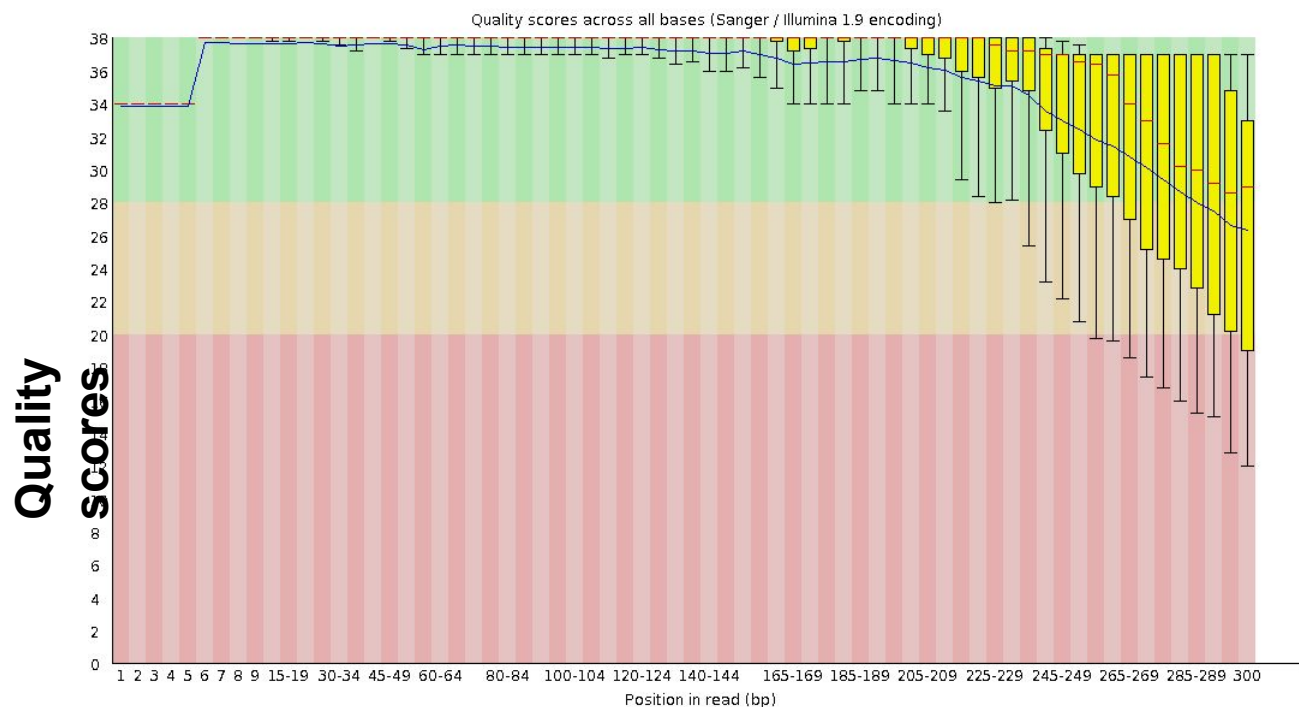
Adapted from the Project Score Cancer Dependency Map

High Throughput Sequencing



[illegible]

Quality scores



Each base has a quality score

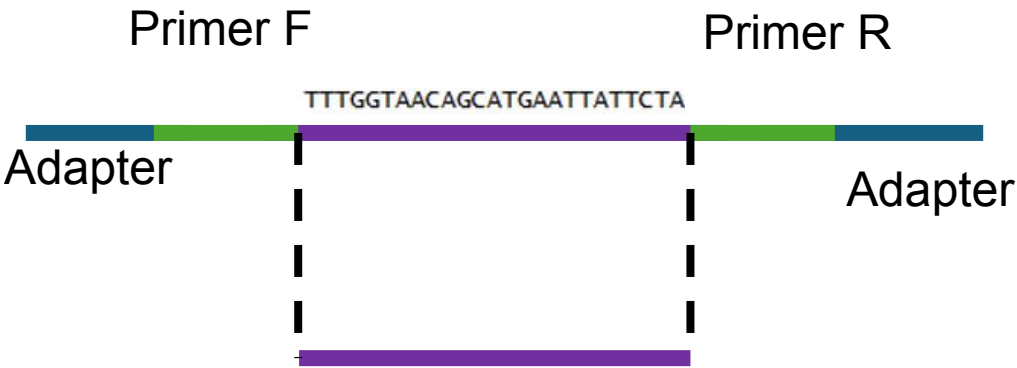
Typically, a decline in base quality moving along the read from 5' -> 3'



Position along read

From reads to counts

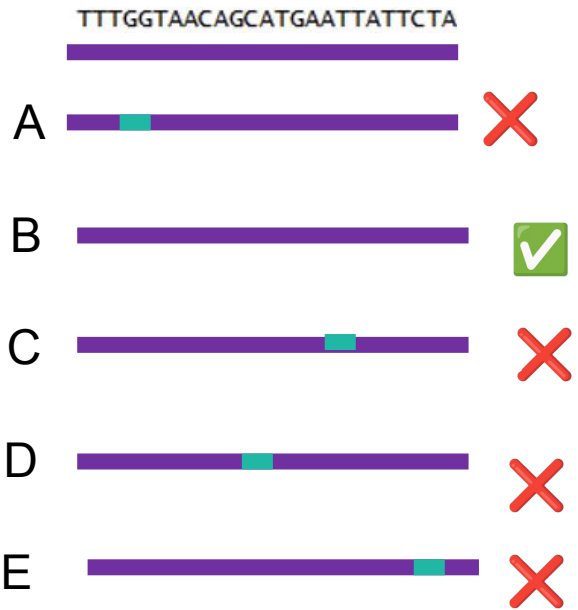
Read processing



Trim adapter sequences from reads to isolate region of interest

Mapping/alignment

Find matches between read and expected features



Tallying

Repeat for all reads to get counts per "feature"

Guide	Counts
A	100
B	120
C	145
D	30
E	150

Guide counting tools

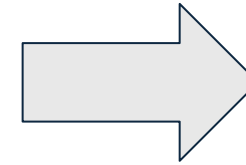
- MAGeCK count

Part of the popular MAGeCK tool

sgRNA ID	Sequence	Gene
s_10007	TGTTACACAGTATAGTTTGCC	CCNA1
s_10008	TTCTCCCTAATTGCTTGCTG	CCNA1
s_10027	ACATGTTGCTTCCCCTTGCA	CCNC

Fastq
.bam
.cram

Guide
seqs



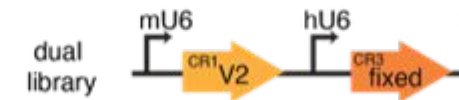
Counts

Guide	Counts
A	100
B	120
C	145
D	30
E	150

- LUCA

Supports multi-guide designs, alternative CRISPR systems and no need to pre-process reads!

```
luca count -o out_dir -l lib_dir exp.yaml reads.bam
```



Analysing high throughput counts

	Replicate 1			Replicate 2			Replicate 3		
	Day 4	Day 7	Day 15	Day 4	Day 7	Day 15	Day 4	Day 7	Day 15
Guide A	100	90	60	102	88	29	102	88	29
Guide B	120	111	112	114	150	100	114	150	100
Guide C	145	40	100	145	42	42	145	42	42
...	...	...	...	...	...	...	...	...	...

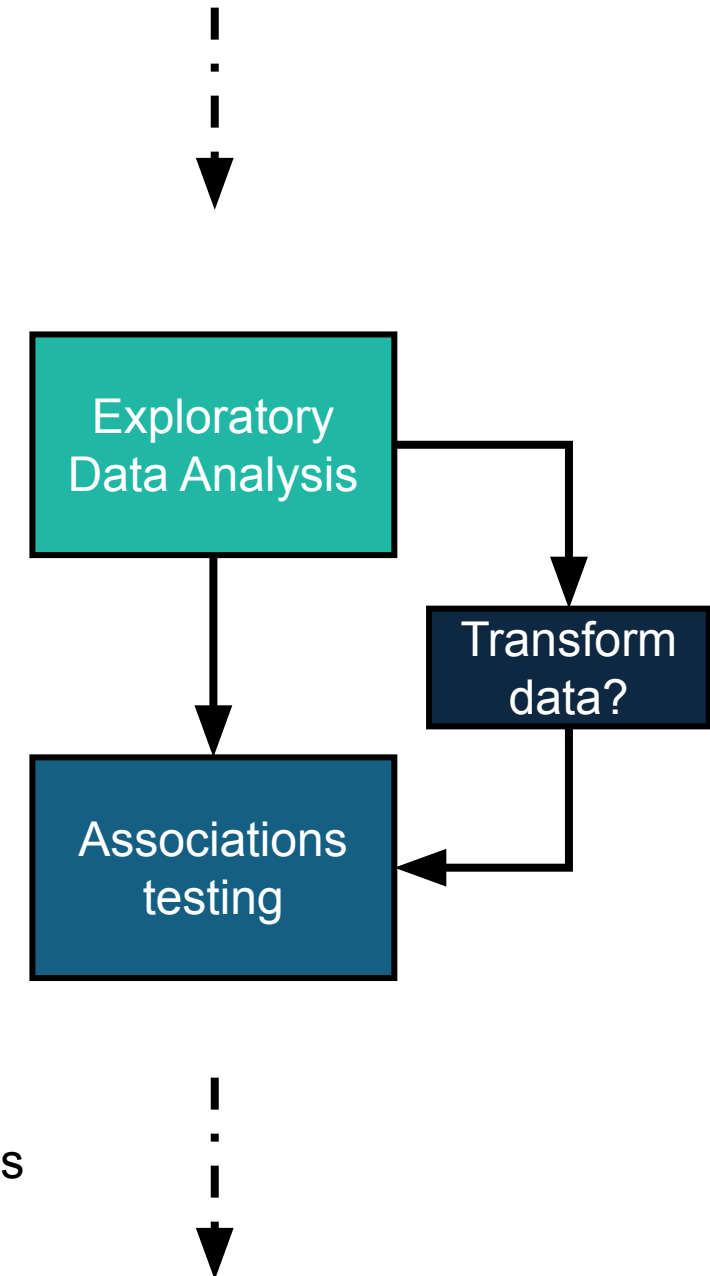
Start from a **counts matrix**: a mathematical object with:

Rows as features (genes, guides, proteins, species of bacteria ...)

Columns as samples (replicates, experimental conditions)

High dimensional (1000's -> 100,000's of features)

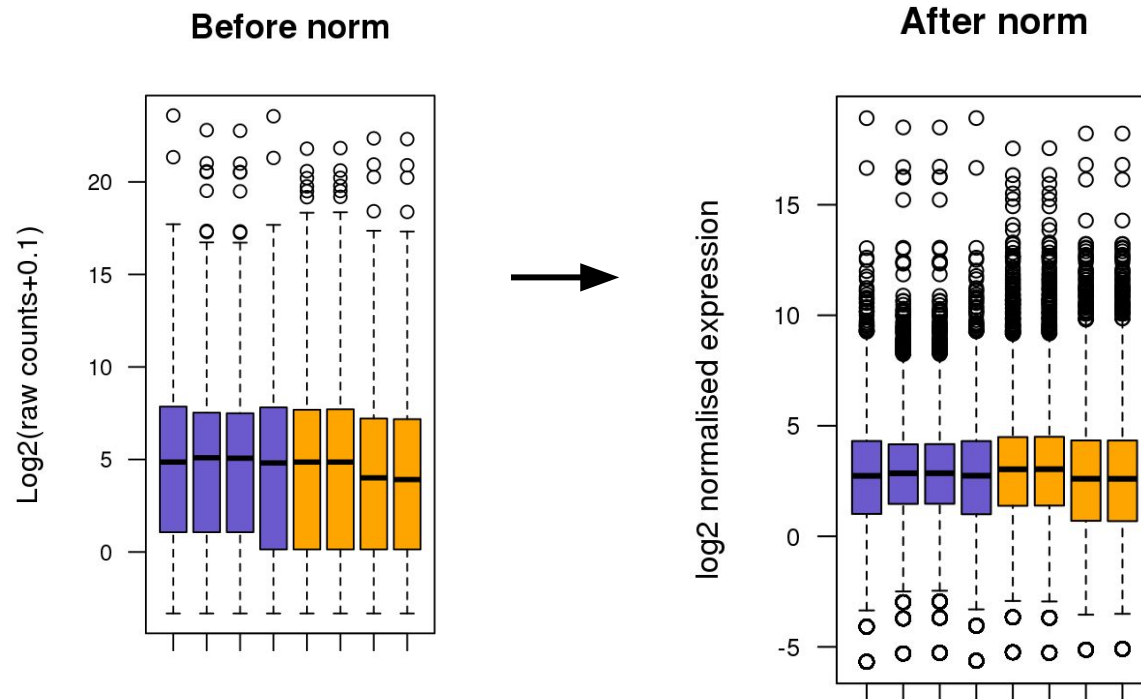
General set of numerical methods for the analysis of different kinds of datasets



Dataset transformation

Normalise to account for technical variability in:

- **Sequencing depth between samples** (1 million in Sample A vs. 2 million reads in Sample B)
- **Sample composition** (highly abundant feature can distort measurements of other features when they change)
- **Feature characteristics** (e.g., gene length in RNA seq)



The diagram illustrates the process of normalizing raw counts. It consists of two tables, one for 'Raw' data and one for 'Normalised counts', with a downward arrow indicating the transformation.

Raw Data Table:

	Replicate 1			Replicate 2			Replicate 3		
	Day 4	Day 7	Day 15	Day 4	Day 7	Day 15	Day 4	Day 7	Day 15
Variant A	100	90	60	102	88	29	102	88	29
Variant B	120	111	112	114	150	100	114	150	100
Variant C	145	40	100	145	42	42	145	42	42
...	...	...	...	...	...	...	...	...	...

Normalised counts Table:

	Replicate 1			Replicate 2			Replicate 3		
	Day 4	Day 7	Day 15	Day 4	Day 7	Day 15	Day 4	Day 7	Day 15
Variant A	100	90	60	102	88	29	102	88	29
Variant B	120	111	112	114	150	100	114	150	100
Variant C	145	40	100	145	42	42	145	42	42
...	...	...	...	...	...	...	...	...	...

The diagram shows that the raw counts are identical to the normalized counts in this specific example, suggesting a normalization factor of 1.0 was applied to all values.

Associations testing

Features whose counts are “associated” with a condition

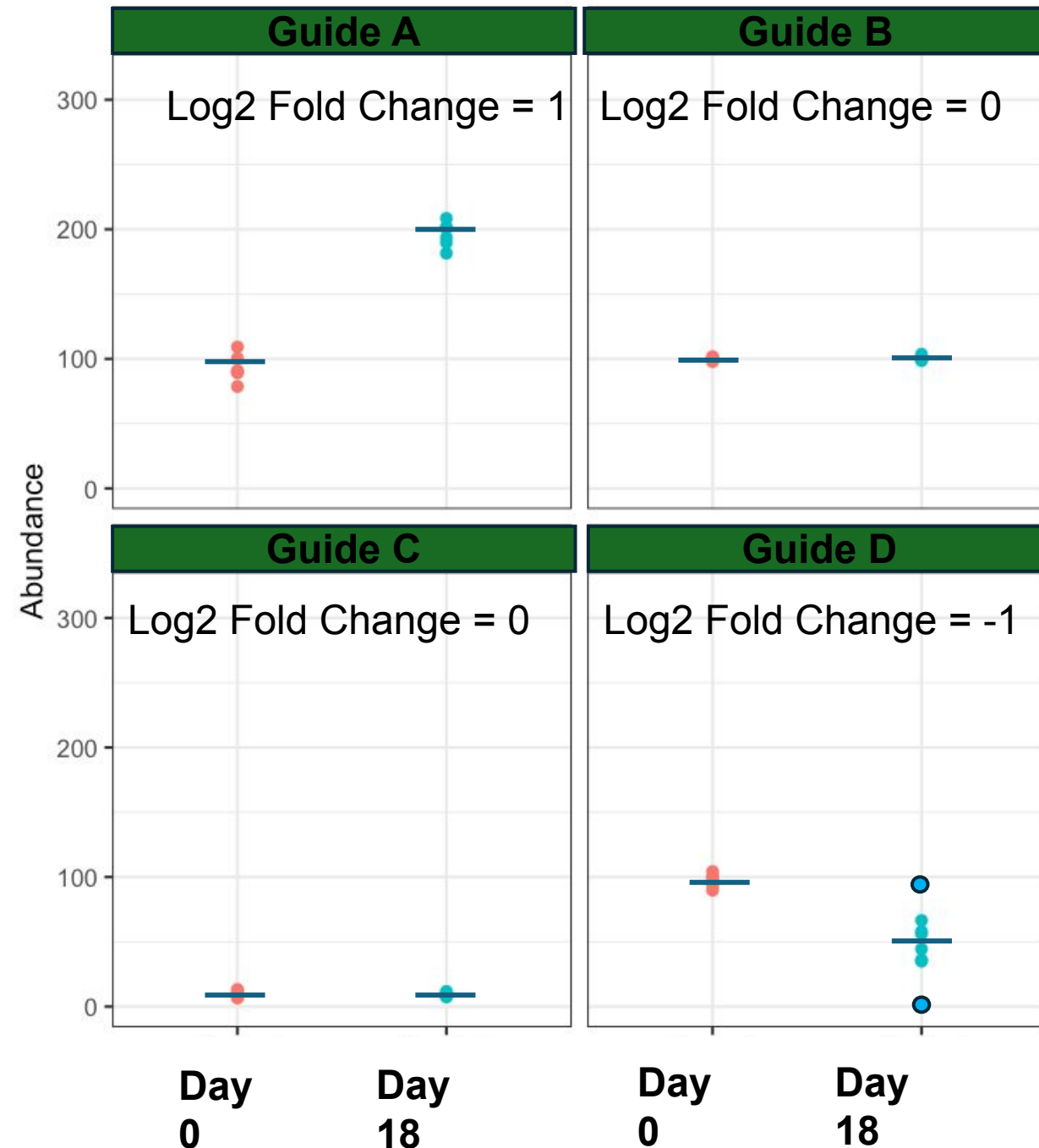
	Replicate 1			Replicate 2			Replicate 3		
	Day 4	Day 7	Day 15	Day 4	Day 7	Day 15	Day 4	Day 7	Day 15
Variant A	100	90	60	102	88	29	102	88	29
Variant B	120	111	112	114	150	100	114	150	100
Variant C	145	40	100	145	42	42	145	42	42
...	...	...	...	...	...	...	...	...	...

Normalised
counts

Guides *associated* with Day 18 timepoint
Proteins *associated* with drug response

Going to perform statistical tests to compare between timepoints of Day 0 vs Day 18

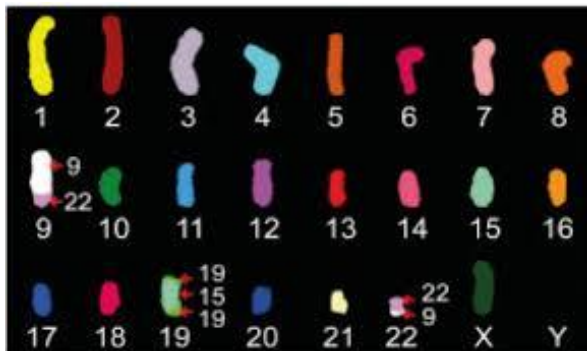
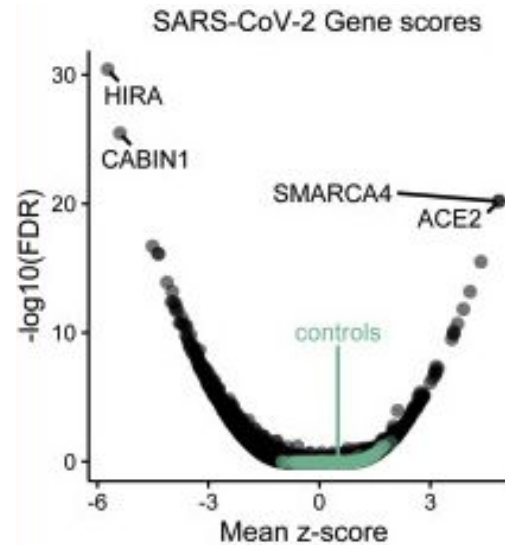
With appropriate modifications for
i) counts data ii) large numbers of comparisons



This practical

Understanding the process of analyzing CRISPR screening data from counts matrix -> gene hits

	Day 0	Day 7	Day 15	Day 0	Day 7	Day 15	Day 0	Day 7	Day 15
Guide A	100	90	60	102	88	29	102	88	29
Guide B	120	111	112	114	150	100	114	150	100
Guide C	145	40	100	145	42	42	145	42	42
...	...	...	...	...	...	...	...	...	...



Finding essential genes in the HAP1 cell line